

Cacti: Cacti is a network graphing solution that uses RRDtool to create visual representations of network traffic and performance metrics.

Usage benefits	How to use it
Graphs bandwidth and network statistics	Configure via web interface
Monitors devices using SNMP	Add devices via the SNMP protocol
Automates data collection and visualisation	Scheduled polling via web interface

By utilising these tools, network administrators can maintain robust network performance and address potential issues before they impact users.

Automation and scripting tools

Automation and scripting tools are essential for network administrators to manage repetitive tasks and streamline configuration processes. These tools save time, reduce errors, and enhance productivity by automating routine operations and ensuring consistent configurations across systems.

Key questions administrators face daily with respect to automation are:

- How can routine tasks be automated to save time and reduce manual effort?
- Are network configurations consistent and efficiently managed across devices?

Task automation

cron: cron is a time-based job scheduling tool in UNIX-like systems that automates repetitive tasks by executing them at specified intervals.

Usage benefits	Example command
Automates routine tasks	<code>crontab -e</code> to edit scheduled jobs
Schedules periodic maintenance tasks	<code>0 2 * * * /usr/bin/backup.sh</code> (daily backup at 2 AM)
Lists current scheduled jobs	<code>crontab -l</code>

at: at schedules one-time tasks to execute at a specific time, providing flexibility for ad-hoc automation.

Usage benefits	Example command
Schedules a one-time job	<code>echo "/usr/bin/update.sh"</code>
Lists pending jobs	<code>atq</code>
Removes a scheduled job	<code>atrm <job_id></code>

Configuration management

Ansible: Ansible is an agentless configuration management tool that automates the deployment and management of systems and applications.

Usage benefits	Example command
Automates system configurations	<code>ansible-playbook site.yml</code>
Deploys applications across multiple servers	<code>ansible all -m copy -a "src=file dest=/path"</code>
Monitors configuration drift	Use Ansible playbooks to enforce consistency

Advanced network tools

Advanced network tools enable administrators to enhance security, manage high availability, and optimise network performance. These tools support VPN management, load balancing, and network discovery, ensuring reliability and scalability.

Key questions administrators face daily in this respect are:

- Is the network secure and accessible for remote users?
- How can traffic be effectively balanced to ensure high availability?

VPN management

OpenVPN: OpenVPN is an open source virtual private network (VPN) solution that provides secure remote access to private networks.

Usage benefits	Example command
Establishes secure connections	<code>openvpn --config client.ovpn</code>
Supports site-to-site VPNs	Configure server and client using .ovpn files
Provides strong encryption and authentication	Use SSL/TLS certificates

WireGuard: WireGuard is a lightweight VPN protocol designed for simplicity and high performance.

Usage benefits	Example command
Creates high-speed VPN connections	<code>wg-quick up wg0</code>
Offers easy configuration	<code>Edit /etc/wireguard/wg0.conf</code>
Ensures low-latency encrypted traffic	Uses modern cryptographic techniques